Education Pathways Into Engineering

The map behind Week 5. Four routes, honestly compared.

Route 1: Community college → transfer (the working person's classic)

Two years of the same calculus, physics, chemistry, and intro engineering at a community college — typically at a third (or less) of university cost — then transfer into a university engineering program for the final two years.

- **Why it fits this plan:** evening/online sections exist, so it stacks with a full-time technician job; tuition reimbursement often covers most or all of community college cost.
- **The one thing you must verify:** a formal **articulation/transfer agreement** between the community college and the target university's *engineering school* (not just the university generally). Engineering programs are picky about which math/physics/intro courses transfer. Meet an advisor at BOTH schools and get the course plan in writing.
- **Watch out for:** taking classes that don't transfer (the advisor meeting prevents this), and the transfer GPA bar for engineering admission.

Route 2: Engineering Technology (MET/EET)

A different degree: **engineering technology** (2-year AS or 4-year BS) is more applied and hands-on, less theoretical math. Graduates become technologists, senior technicians, manufacturing/test/field engineers.

- **Honest trade-off:** a BS in Engineering Technology is *not* interchangeable with a BS in Engineering. Some employers treat them the same for many roles; some don't. Licensure (PE) with an ET degree is possible in some states, harder in others. If the dream is "design engineer at an aerospace firm," lean Route 1/4. If it's "well-paid hands-on technical career, working sooner," ET is excellent and *underrated*.
- **Fits working students well** — programs are often at community colleges and regional universities with evening options, and coursework connects directly to a technician day job.

Route 3: Apprenticeship (earn from day one)

Paid work + paid classroom training → journeyman credential in ~4-5 years, no debt, solid income the whole way. Electrician (IBEW), machinist, industrial maintenance, mechatronics.

- **Honest trade-off:** it's a trades credential, not an engineering degree — a different (excellent) career, not a slower route to the same one. Plenty of people later stack a degree on top (some locals/employers help pay), and journeyman experience makes a later EE/ME student dramatically better at the job.

- **Reality check:** good apprenticeships are competitive with application windows and aptitude tests (algebra matters). Missing a window can mean waiting a year — find the dates *early*.

Route 4: Straight to a 4-year university

The default path. Four (realistically often five) years, highest sticker price, full college experience, engineering internships between years.

- **When it wins:** strong financial aid, scholarships, or family funding; or admission to a program whose brand/co-op pipeline genuinely changes outcomes.
- **When it loses:** paying university prices for the freshman/sophomore courses community college sells for a third the price, funded by loans instead of an employer.

ABET, in one paragraph

ABET accreditation is the quality stamp for engineering and engineering technology programs. It matters because: (1) the FE/PE licensure track expects an ABET-accredited degree, (2) many employers (especially defense) require it, and (3) transfer agreements are built around it. Before committing money to ANY program, search it on abet.org. Community colleges themselves usually aren't ABET-accredited (that's normal — the *destination* program is the one that must be), but pre-engineering courses must be transferable to an ABET program.

The combination most likely to fit this plan

Technician job (full-time) + community college (evenings) + employer tuition money → transfer or ET completion. Slower per-year than full-time school; graduates at ~23-24 with experience, income history, industry contacts, and little or no debt — usually *ahead* of the classmate who went straight through, in everything except graduation date. But Week 5's job is to check whether that's true *for you, where you live* — not to take this file's word for it.